

Risk Stratification and Subgroup Patterns With Cardionex in Adults With Chronic Heart Failure: A Simulated Biomarker-Informed Analysis for Clinical Management

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Abstract

Chronic heart failure is clinically heterogeneous, and treatment response may vary according to baseline functional limitation, comorbidity burden, renal status, prior hospitalization history, and biomarker-defined risk patterns. Cardionex, the drug name associated with Corvanta, is a simulated investigational therapeutic concept developed for internal scientific-content workflow demonstration in chronic heart failure management. This simulated biomarker-informed subgroup analysis evaluated whether baseline clinical-risk profiles were associated with differential symptom, functional, quality-of-life, and hospitalization-related outcomes among adults receiving Cardionex-based care. A total of 984 simulated adults with chronic heart failure were followed for 40 weeks. Participants were stratified by age, baseline functional status, renal-function category, prior hospitalization history, comorbidity burden, congestion-risk profile, and natriuretic-peptide risk category. Outcomes included changes in symptom burden, 6-minute walk distance, quality-of-life limitation, outpatient care escalation, and heart failure hospitalization. Greater simulated improvements were observed among participants with moderate baseline functional limitation, elevated but not extreme congestion-risk profiles, preserved monitoring eligibility, and prior hospitalization history without advanced instability. Participants with severe renal limitation, high comorbidity burden, and advanced functional impairment showed smaller gains and higher care-escalation rates. These findings suggest that risk stratification may help identify patient profiles most suitable for structured Cardionex-based management. All data are simulated and intended only for internal evidence mapping, subgroup-narrative testing, and medical-affairs workflow demonstration.

Keywords: Chronic heart failure; Cardionex; Corvanta; risk stratification; subgroup analysis; biomarker-informed care; renal function; hospitalization risk

Introduction

Chronic heart failure is not a single uniform condition. Patients differ substantially in symptom severity, functional status, comorbidity burden, renal function, prior hospitalization history, background therapy patterns, congestion risk, and ability to complete structured monitoring. These differences can influence treatment persistence, clinical stability, safety monitoring, care escalation, and patient-reported outcomes.

For cardiologists and internal medicine specialists, risk stratification is essential because chronic heart failure management often requires individualized clinical decisions. A patient with moderate symptoms, stable renal function, and a recent hospitalization history may require a different management strategy than a patient with advanced functional limitation, severe renal instability, frequent decompensation, or high frailty burden. Understanding which patient profiles are most likely to show favorable management outcomes can support better monitoring design, content development, and evidence communication.

Subgroup and biomarker-informed analyses are especially useful in chronic heart failure because outcomes are shaped by baseline risk. Biomarkers and clinical-risk markers can help contextualize symptom trajectory, functional capacity, congestion status, and hospitalization risk. However, subgroup findings must be interpreted cautiously because they are frequently exploratory and may not establish causal treatment-response differences.

Corvanta is the brand name associated with Cardionex, a simulated investigational therapeutic concept created for internal scientific-content workflow demonstration. Cardionex is positioned as a chronic heart failure management concept evaluated within structured care pathways, with attention to symptom burden, functional capacity, patient stability, renal monitoring, and care escalation.

The objective of this simulated biomarker-informed subgroup analysis was to evaluate whether baseline patient characteristics were associated with differential clinical-management outcomes among adults receiving Cardionex-based care for chronic heart failure.

Methods

Study Design

This was a simulated, prospective, biomarker-informed subgroup analysis conducted over 40 weeks. The analysis evaluated adults with chronic heart failure receiving Cardionex-based care within a structured chronic heart failure management framework. Subgroups were defined using baseline clinical characteristics, care history, comorbidity burden, renal-function category, congestion-risk profile, and natriuretic-peptide risk category.

This manuscript contains simulated data generated for internal scientific-content workflow demonstration. The design, population, endpoints, subgroup categories, and results are hypothetical and should not be interpreted as real-world clinical evidence, product efficacy, product safety, regulatory support, guideline endorsement, or prescribing guidance for Corvanta or Cardionex.

Study Population

The simulated analysis included adults with chronic heart failure who had persistent symptoms despite background care and sufficient baseline clinical-risk documentation for subgroup classification. The population was designed to reflect a global chronic heart failure management setting, including cardiology and internal medicine practice environments.

Inclusion Criteria

Participants were eligible for inclusion if they met all of the following simulated criteria:

- Adults aged 40 years or older.
- Documented chronic heart failure for at least 6 months.
- Persistent symptom burden requiring structured follow-up.
- Initiation of Cardionex-based care within a simulated chronic heart failure management pathway.
- Baseline assessment available for functional status, renal function, comorbidity burden, and hospitalization history.
- Availability of at least one baseline biomarker-informed risk category.
- Ability to complete symptom and functional assessments during follow-up.

Exclusion Criteria

Participants were excluded if they met any of the following simulated criteria:

- Acute decompensated heart failure requiring hospitalization at enrollment.
- Planned advanced heart failure intervention during the first 12 weeks.
- Severe unstable arrhythmia requiring urgent intervention.
- Rapidly progressive renal dysfunction preventing structured monitoring.
- Incomplete baseline subgroup classification.
- Severe non-cardiac illness expected to dominate outcomes.
- Inability to complete follow-up symptom or functional-capacity assessments.

Intervention or Exposure Concept

Participants received Cardionex-based care as part of a simulated structured chronic heart failure management pathway. The pathway included symptom review, functional assessment, background-care evaluation, renal and electrolyte monitoring, congestion-status education, follow-up scheduling, and care-escalation review.

Cardionex exposure in this manuscript is a simulated investigational concept. No dosing, prescribing, or product-use instruction should be inferred from this document.

Subgroup Definitions

The following baseline subgroups were evaluated:

- Age group: <65 years, 65–74 years, and ≥ 75 years.

- Baseline functional limitation: mild, moderate, or advanced.
- Prior heart failure hospitalization within 12 months: yes or no.
- Renal-function category: preserved, mildly reduced, moderately reduced, or severely reduced.
- Comorbidity burden: low, moderate, or high.
- Congestion-risk profile: low, intermediate, or high.
- Natriuretic-peptide risk category: lower, intermediate, or elevated.
- Background therapy complexity: standard, complex, or highly complex.

Endpoints

The primary endpoint was change from baseline to week 40 in Heart Failure Symptom Burden Score, a simulated 0–100 scale where higher scores indicate greater symptom burden.

Secondary endpoints included:

- Change from baseline in 6-minute walk distance.
- Change from baseline in Heart Failure Quality-of-Life Limitation Score.
- Proportion of participants with clinically meaningful symptom improvement.
- Outpatient care escalation for worsening symptoms or congestion concern.
- Urgent care visit for worsening heart failure symptoms.
- Heart failure hospitalization through week 40.
- Treatment-emergent adverse events and monitoring alerts.

Statistical Analysis

The simulated analysis included all participants with baseline subgroup data and at least one follow-up assessment. Continuous outcomes were summarized using means, standard deviations, and mean changes from baseline. Categorical outcomes were summarized using counts and percentages. Subgroup patterns were interpreted descriptively. No formal causal inference should be assumed from these simulated results.

Exploratory interaction patterns were summarized using descriptive differences across baseline categories. Missing data were handled using simulated repeated-measure summaries and observed-case sensitivity checks.

Ethics and Compliance Statement

This simulated analysis was reviewed under a fictional internal research-governance process designed for scientific-content workflow demonstration. No real patients, clinical sites, investigators, ethics committees, trial registries, regulatory authorities, or real-world datasets were involved. All data, analyses, subgroup categories, and author affiliations are fictional and were created solely for internal medical-affairs workflow testing.

Results

Participant Disposition and Baseline Risk Profile

A total of 984 simulated participants were included in the analysis. Week 40 follow-up data were available for 891 participants. Baseline risk categories showed a clinically heterogeneous population with meaningful variation in functional limitation, renal function, prior hospitalization history, and comorbidity burden.

Baseline Characteristic	Overall Cohort (N = 984)
Mean age, years	68.2 ± 9.6
Age <65 years	316 (32.1%)
Age 65–74 years	401 (40.8%)
Age ≥75 years	267 (27.1%)
Female participants	417 (42.4%)
Prior heart failure hospitalization within 12 months	336 (34.1%)
Moderate baseline functional limitation	542 (55.1%)
Advanced baseline functional limitation	218 (22.2%)
Moderately reduced renal-function category	296 (30.1%)
Severely reduced renal-function category	86 (8.7%)
High comorbidity burden	331 (33.6%)
High congestion-risk profile	271 (27.5%)
Elevated natriuretic-peptide risk category	309 (31.4%)

Overall Clinical-Management Outcomes

At week 40, the mean Heart Failure Symptom Burden Score improved by –12.9 points from baseline. Mean 6-minute walk distance increased by 36 meters, and quality-of-life limitation improved by –10.4 points. Heart failure hospitalization occurred in 9.9% of participants, while outpatient care escalation occurred in 22.5%.

Overall Outcome at Week 40	Simulated Result (N = 984)
Mean change in symptom burden score	–12.9 ± 13.8
Participants with ≥10-point symptom improvement	548 (55.7%)
Mean change in 6-minute walk distance	+36 ± 66 meters
Mean change in quality-of-life limitation score	–10.4 ± 14.1

Overall Outcome at Week 40	Simulated Result (N = 984)
Outpatient care escalation	221 (22.5%)
Urgent care visit for worsening heart failure symptoms	137 (13.9%)
At least one heart failure hospitalization	97 (9.9%)
Treatment-emergent adverse event	476 (48.4%)
Monitoring alert requiring care review	186 (18.9%)

Subgroup Outcomes by Baseline Functional Limitation

Participants with moderate baseline functional limitation showed the greatest simulated improvement in symptom burden, functional capacity, and quality-of-life limitation. Participants with advanced baseline functional limitation also improved, but gains were smaller and care-escalation rates were higher.

Baseline Functional Limitation	n	Symptom Burden Change	6-Minute Walk Distance Change	Quality-of-Life Limitation Change	Heart Failure Hospitalization
Mild	224	-10.6 ± 12.7	+29 ± 58 m	-8.7 ± 12.8	14 (6.3%)
Moderate	542	-15.2 ± 13.4	+44 ± 64 m	-12.5 ± 13.6	42 (7.7%)
Advanced	218	-8.4 ± 14.5	+21 ± 73 m	-6.8 ± 15.2	41 (18.8%)

Subgroup Outcomes by Prior Hospitalization History

Participants with a prior heart failure hospitalization had higher baseline risk but showed meaningful simulated improvement in symptoms and quality-of-life limitation. Despite improvement, hospitalization and care-escalation rates remained higher in this subgroup compared with participants without recent hospitalization history.

Prior Heart Failure Hospitalization Within 12 Months	n	Symptom Burden Change	Quality-of-Life Limitation Change	Outpatient Care Escalation	Heart Failure Hospitalization
Yes	336	-14.1 ± 14.2	-11.6 ± 14.7	96 (28.6%)	49 (14.6%)
No	648	-12.3 ± 13.5	-9.7 ± 13.8	125 (19.3%)	48 (7.4%)

Subgroup Outcomes by Renal-Function Category

Participants with preserved or mildly reduced renal-function categories had more favorable simulated symptom, functional, and safety-monitoring outcomes. Participants with severely reduced renal-function category showed smaller improvements and a higher frequency of renal or electrolyte monitoring alerts.

Renal-Function Category	n	Symptom Burden Change	6-Minute Walk Distance Change	Monitoring Alert Requiring Care Review	Discontinuation Due to Safety/Tolerability
Preserved	251	-14.8 ± 12.9	+43 ± 61 m	28 (11.2%)	6 (2.4%)
Mildly reduced	351	-13.7 ± 13.1	+39 ± 64 m	52 (14.8%)	13 (3.7%)
Moderately reduced	296	-11.5 ± 14.4	+29 ± 69 m	74 (25.0%)	19 (6.4%)
Severely reduced	86	-6.9 ± 15.1	+14 ± 72 m	32 (37.2%)	11 (12.8%)

Subgroup Outcomes by Comorbidity Burden and Congestion-Risk Profile

Participants with moderate comorbidity burden and intermediate congestion-risk profile showed consistent simulated improvements while maintaining manageable care-escalation rates. Participants with high comorbidity burden and high congestion-risk profile had smaller outcome gains and more frequent urgent-care use.

Risk Category	n	Symptom Burden Change	Quality-of-Life Limitation Change	Urgent Care Visit	Heart Failure Hospitalization
Low comorbidity burden	258	-13.1 ± 12.8	-10.7 ± 13.1	24 (9.3%)	14 (5.4%)
Moderate comorbidity burden	395	-14.3 ± 13.5	-11.8 ± 13.9	43 (10.9%)	30 (7.6%)
High comorbidity burden	331	-10.9 ± 14.6	-8.2 ± 15.0	70 (21.1%)	53 (16.0%)
Low congestion-risk profile	303	-11.2 ± 12.4	-8.9 ± 12.9	28 (9.2%)	17 (5.6%)
Intermediate congestion-risk profile	410	-15.1 ± 13.7	-12.6 ± 14.0	49 (12.0%)	31 (7.6%)
High congestion-risk profile	271	-10.3 ± 15.0	-7.8 ± 15.4	60 (22.1%)	49 (18.1%)

Biomarker-Informed Risk Category Outcomes

Participants in the intermediate natriuretic-peptide risk category showed the most consistent simulated improvement in symptom burden and functional capacity. Participants in the elevated category remained at higher risk of hospitalization and care escalation despite symptom improvement.

Natriuretic-Peptide Risk Category	n	Symptom Burden Change	6-Minute Walk Distance Change	Outpatient Care Escalation	Heart Failure Hospitalization
Lower	287	-10.8 ± 12.6	+31 ± 59 m	43 (15.0%)	15 (5.2%)
Intermediate	388	-15.4 ± 13.2	+45 ± 63 m	74 (19.1%)	28 (7.2%)
Elevated	309	-11.1 ± 15.1	+24 ± 73 m	104 (33.7%)	54 (17.5%)

Safety and Monitoring by Risk Category

Safety and monitoring outcomes varied by baseline risk. Monitoring alerts were more frequent among participants with severely reduced renal-function category, high comorbidity burden, and highly complex background therapy. Most monitoring alerts were managed through protocol-defined care review, temporary interruption, background-care adjustment, hydration review, or repeat laboratory assessment within the simulated framework.

Safety or Monitoring Outcome	Lower-Risk Composite Group (n = 286)	Intermediate-Risk Composite Group (n = 462)	Higher-Risk Composite Group (n = 236)
Any treatment-emergent adverse event	112 (39.2%)	213 (46.1%)	151 (64.0%)
Dizziness	22 (7.7%)	43 (9.3%)	34 (14.4%)
Symptomatic hypotension	9 (3.1%)	20 (4.3%)	19 (8.1%)
Renal-function monitoring alert	18 (6.3%)	47 (10.2%)	51 (21.6%)
Electrolyte monitoring alert	11 (3.8%)	35 (7.6%)	38 (16.1%)
Serious adverse event	18 (6.3%)	41 (8.9%)	39 (16.5%)
Discontinuation due to safety/tolerability	7 (2.4%)	20 (4.3%)	22 (9.3%)

Discussion

This simulated biomarker-informed subgroup analysis suggests that baseline clinical-risk profiles may help identify adults with chronic heart failure who are more likely to show favorable symptom, functional, and quality-of-life outcomes during Cardionex-based structured care. The most consistent simulated improvements were observed among participants with moderate baseline functional limitation, intermediate congestion-risk profile, intermediate natriuretic-peptide risk category, preserved or mildly reduced renal function, and manageable comorbidity burden.

The finding that participants with moderate functional limitation showed the greatest improvement is clinically plausible within a simulated framework. Patients with mild limitation may have less room for measurable improvement, while patients with advanced limitation may have higher competing risks, more fixed functional impairment, and greater comorbidity burden. This pattern can support internal medical-affairs messaging around appropriate patient characterization without implying a confirmed real-world treatment-response profile.

Prior hospitalization history identified a subgroup with higher baseline risk. Participants with a recent hospitalization history showed meaningful symptom and quality-of-life improvements but continued to experience higher hospitalization and care-escalation rates than those without recent hospitalization. This reinforces the concept that improvement in patient-centered outcomes does not necessarily eliminate residual clinical risk. For cardiologists and internal medicine specialists, this distinction is important when discussing chronic heart failure stability and ongoing monitoring needs.

Renal-function category was a major differentiator of simulated safety and monitoring outcomes. Participants with preserved or mildly reduced renal function showed more favorable symptom and functional outcomes and fewer monitoring alerts, while those with severely reduced renal-function category had smaller gains and higher rates of renal or electrolyte monitoring alerts. This supports a balanced internal narrative emphasizing renal monitoring, individualized care review, and careful interpretation of outcomes among higher-risk patients.

Comorbidity burden and congestion-risk profile also shaped the simulated results. Participants with moderate comorbidity burden and intermediate congestion-risk profile appeared to have the most favorable combination of improvement potential and manageable care escalation. In contrast, high comorbidity burden and high congestion-risk profile were associated with higher urgent-care and hospitalization rates. These findings may help structure internal evidence maps that separate symptom improvement, risk persistence, monitoring burden, and care coordination needs.

The biomarker-informed findings should be interpreted descriptively. Intermediate natriuretic-peptide risk category was associated with greater simulated functional and symptom improvement, while elevated risk category remained associated with higher care escalation and hospitalization. This does not establish a validated biomarker-response relationship. Instead, it demonstrates how a medical-affairs evidence workflow may organize patient segments for scientific discussion, claim review, and content-development testing.

Overall, this simulated analysis highlights the value of subgroup thinking in chronic heart failure management. It supports the development of balanced scientific content that recognizes patient heterogeneity, residual risk, monitoring needs, and the importance of matching communication to clinically relevant patient profiles.

Conclusion

In this simulated biomarker-informed subgroup analysis of adults with chronic heart failure, Cardionex-based structured care was associated with variable outcomes across baseline clinical-risk groups. The most favorable simulated patterns were observed among participants with moderate functional limitation, intermediate congestion-risk profile, intermediate natriuretic-peptide risk category, preserved or mildly reduced renal function, and manageable comorbidity burden. Participants with advanced functional

limitation, severe renal-function reduction, high comorbidity burden, and high congestion-risk profile had smaller improvements and higher care-escalation or monitoring-alert rates.

These findings are simulated and should be used only for internal scientific-content workflow demonstration, subgroup-narrative testing, evidence mapping, and medical-affairs planning. They should not be interpreted as real-world clinical evidence, validated patient-selection criteria, regulatory support, guideline endorsement, or prescribing guidance for Corvanta or Cardionex.

Medical Affairs Interpretation

This simulated research paper can support internal medical-affairs workflows by providing a structured subgroup-analysis asset for chronic heart failure content development. It may help teams test how patient heterogeneity, baseline risk, renal monitoring, and residual hospitalization risk are represented in scientific narratives.

Potential internal applications include:

- **Subgroup evidence mapping:** The subgroup tables can support mapping of functional limitation, renal status, hospitalization history, comorbidity burden, and biomarker-informed risk categories.
- **Patient-profile narrative development:** Findings can help structure balanced descriptions of which patient profiles may show greater simulated improvement versus those requiring closer monitoring.
- **Claim-source alignment testing:** The paper can be used to test whether generated claims remain appropriately qualified and avoid overstating subgroup conclusions.
- **MLR pre-screen demonstration:** The manuscript can support review of subgroup language, risk-based claims, biomarker references, and safety-monitoring statements.
- **HCP content planning:** The analysis may inform congress posters, medical slide decks, field medical summaries, HCP discussion guides, and risk-segmentation educational assets.
- **Care-pathway workflow testing:** Subgroup patterns can help demonstrate how DAWN-style content workflows may connect patient profiles to monitoring, care escalation, and evidence interpretation.

Any downstream content should clearly preserve the simulated nature of the evidence and avoid implying confirmed clinical benefit, validated patient selection, regulatory approval, or guideline endorsement.

Limitations

This manuscript has several important limitations. First, all data are simulated and do not represent real clinical outcomes for Corvanta or Cardionex. Second, subgroup definitions and biomarker-informed categories were created for internal workflow demonstration and have not been validated. Third, the analysis is descriptive and should not be interpreted as causal evidence of differential treatment response. Fourth, baseline risk factors such as renal function, comorbidity burden, congestion status, and prior hospitalization history are interrelated, making interpretation of individual subgroup patterns complex. Fifth, safety and monitoring findings are hypothetical and should not be used to infer the actual tolerability profile of any real product.

Because of these limitations, the manuscript should be used only as a controlled internal scientific-content asset for workflow testing, evidence structuring, subgroup-narrative development, and medical-affairs demonstration.

Data Transparency Statement

All study data, participant characteristics, subgroup definitions, biomarker-informed categories, endpoints, outcomes, safety findings, author names, affiliations, governance statements, and references in this manuscript are simulated for internal scientific-content workflow demonstration. No real patients, investigators, clinical sites, ethics committees, regulatory authorities, trial identifiers, journal records, DOIs, PMIDs, or real-world datasets are associated with this manuscript. The content should not be interpreted as real-world clinical evidence, validated patient-selection guidance, regulatory support, guideline endorsement, or prescribing guidance for Corvanta or Cardionex.

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